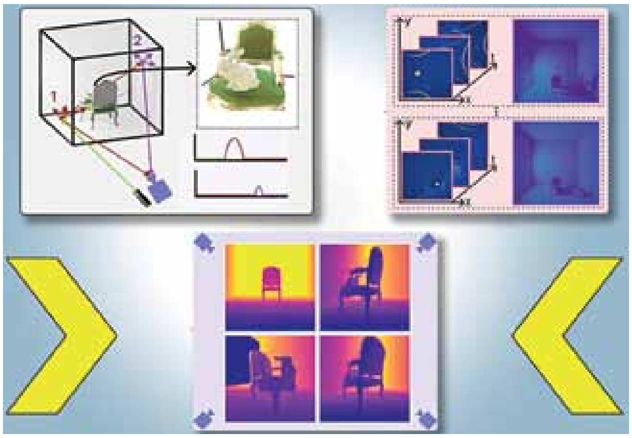
Self-driving cars with new vision tech

Researchers from MIT and Meta have developed PlatoNeRF, a computer vision technique that allows autonomous vehicles to detect obstacles beyond their direct line of sight by using shadows to create detailed 3D models of obscured areas. This



PlatoNeRF is a computer vision system that combines lidar measurements with machine learning to reconstruct a 3D scene, including hidden objects, from only one camera view by exploiting shadows. Here, the system accurately models the rabbit in the chair, even though that rabbit is blocked from view (Credit: Courtesy of the researchers, edited by MIT News)

method combines single photon lidar with machine learning to improve 3D reconstructions, especially in challenging lighting conditions. PlatoNeRF is beneficial for applications such as enhancing AR/VR headsets and aiding warehouse robots in precise environment modelling. By capturing scattered light and analysing secondary light rays, it can infer the geometry of hidden objects. The system integrates multi-bounce lidar with a neural radiance field (NeRF), a machine-learning model that enhances the ability to estimate novel scene views. PlatoNeRF outperforms traditional methods that use only lidar or a NeRF with a colour image, particularly when using lower-resolution sensors.

AI-powered camera tech to detect drunk drivers

Researchers at Edith Cowan University (ECU), in collaboration with Mix by Powerfleet, have developed computer tracking technology to detect alcohol impairment in drivers using standard RGB camera footage. The system analyses facial features, gaze direction, and head position from videos of drivers with different intoxication levels. Tested in a controlled simulator, the technology achieved 75% accuracy in classifying intoxication into three levels. This innovation could be integrated into roadside surveillance, aiding law enforcement in preventing drunk driving. This research highlights the potential for broader applications, including driver monitoring systems and smartphones.

Autonomous water extractor for arid climates

Researchers at Southern Federal University (SFedU) in Russia have introduced an autonomous water extraction device for arid climates. It utilises a metal-organic framework polymer

The prototype installation created by SFedU scientists (Credit: https://www.ismatimes.com)

sorbent to absorb water vapour at night and release it using solar energy during the day. Undergoing field tests in southern Russia, the device aims to address water shortages in regions lacking infrastructure, like parts of Africa and Southeast Asia. Developed by SFedU's International Laboratory of New Educational Technologies, led by Ilya Pankin, the technology is cost-effective, producing water at 9-10 rubles per litre ($0.12-0.14). It operates independently on solar power and functions effectively in humidity levels below 25%.

Thermal switching materials for lithium-ion batteries

Researchers at Tsinghua University and Zhejiang University have developed a new thermal-switching material that quickly responds to temperature changes to regulate heat within high-capacity batteries, as detailed in a Nature Energy paper. This material consists of microspheres embedded between graphene layers, which expand with temperature shifts, disrupting heat transport by separating the graphene layers. This action helps maintain battery cell temperature and prevents explosions. The team tested this material in a 50Ah Ni-Co-Mn lithium-ion battery as an interlayer, where it successfully regulated heat and prevented explosive chain reactions. This thermal regulator shows promise for use in other high-capacity batteries, potentially aiding their safe commercialisation across various climates and operational conditions. The material switches effectively between conducting and insulating states, maintaining uniform temperature distribution and blocking 80% of heat transmission during thermal runaway.